Bayesian quantile-based correction and synthesis for climate products

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QUESTION
Correct first, then synthesize

Can quantile-level bias be learned from diverse climate-product archives and propagated into newly issued ensemble forecasts on operational update timescales, while preserving an interpretable dynamic decomposition of source corrections, trend, seasonality, and exogenous effects?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge on the $\log(1+x)$ scale. Before each forecast origin, observed flow is paired with retrospective ECMWF/GloFAS and NOAA/NWS products to estimate source corrections. After the origin, issued ensembles and staged exogenous covariates are used to carry those corrections into source-adjusted quantile forecasts and one posterior predictive distribution.

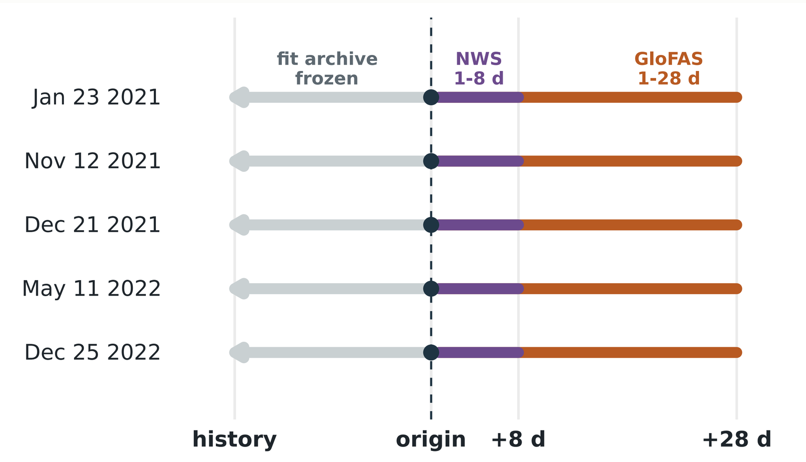
USGS observations
15-min public gauge record aggregated to daily log-flow; state-updating and held-out verification target.

ECMWF / GloFAS
European Centre for Medium-Range Weather Forecasts; retrospectives; daily 51-member ensemble forecasts, leads 1–28.

NOAA / NWS
U.S. National Weather Service under NOAA; retrospectives; hourly 7-member ensemble forecasts, daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION
Rolling-origin holdout design



Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE
Scalable Bayesian quantile fitting

Seven independent extended Dynamic Quantile Linear Model (exDQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. Posterior predictive samples are generated for each fitted quantile lane and synthesized into one posterior predictive distribution.

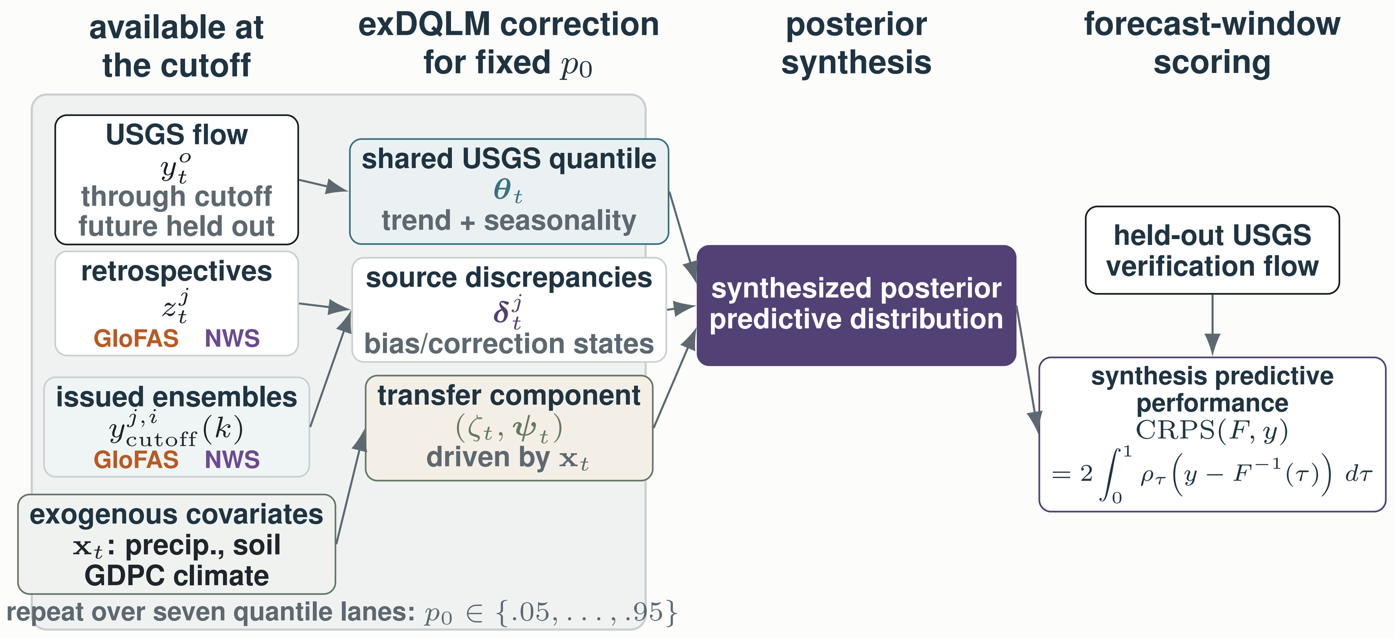
7	$\leq$12,995	$\sim$2–4 h
quantile lanes	USGS archive days	fit + synthesis

Parallel quantile lanes
Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

DQLM baseline versus exDQLM
The asymmetric-Laplace DQLM is conjugate under its latent representation; the selected exDQLM replaces AL by exAL for source-specific scale and skewness flexibility.

Scalable VB-LD updates
Mean-field variational Bayes keeps Kalman filtering and backward smoothing for Gaussian state blocks; VB-LD uses local modes, curvature, and Delta-method moments for non-conjugate exAL terms.

MODEL
Main Workflow



Source-aware exDQLM workflow. For each target quantile p_0, pre-origin USGS observations and retrospective product histories identify a latent river-flow quantile and source-specific correction states. After the cutoff, issued ensemble forecasts and staged exogenous covariates propagate corrected forecast-window quantiles. The seven fitted quantile lanes are then synthesized into one posterior predictive distribution and scored only with held-out USGS observations.

Dynamic Quantile Linear Model (exDQLM)
For each target quantile p_0, let T denote the forecast origin/cutoff. The selected exDQLM links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Source discrepancies learn pre-origin corrections; forecast ensembles and staged covariates then propagate corrected quantiles after the origin.

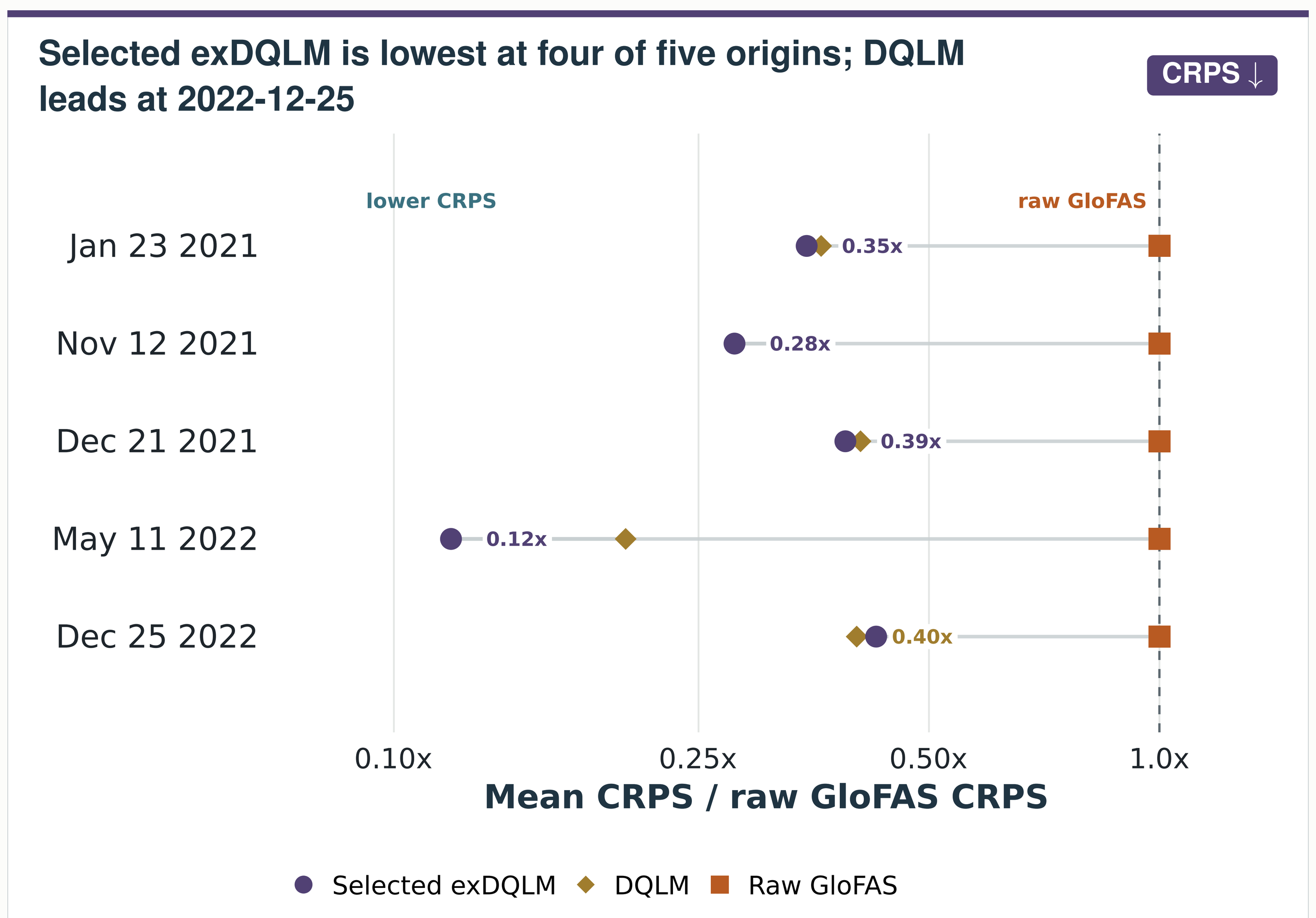
$$
\begin{aligned}
y_t^o &\sim \mathrm{exAL}_{p_0}(\mathbf{F}_t'\boldsymbol{\theta}_t + \zeta_t, \sigma^o, \gamma^o), && 1 \le t \le T,\\
z_t^j &\sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_t'(\boldsymbol{\theta}_t + \boldsymbol{\delta}_t^j) + \zeta_t, \sigma^j, \gamma^j\right), && j \in \mathcal{J},\ 1 \le t \le T,\\
y_T^{j,i}(k) &\sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_{T+k}'(\boldsymbol{\theta}_{T+k} + \boldsymbol{\delta}_{T+k}^j) + \zeta_{T+k}, \sigma^j, \gamma^j\right), && j \in \mathcal{J},\ i \in \mathcal{I}_j,\ k \in \mathcal{K}_j.
\end{aligned}
$$

$$
\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t^{\theta}), \qquad \boldsymbol{\delta}_t^j \mid \boldsymbol{\delta}_{t-1}^j \sim \mathcal{N}(\mathbf{G}_t\boldsymbol{\delta}_{t-1}^j, \mathbf{W}_t^{\delta^j}).
$$

$$
\zeta_t \mid \zeta_{t-1}, \boldsymbol{\psi}_{t-1} \sim \mathcal{N}(\lambda\zeta_{t-1} + \mathbf{x}_t'\boldsymbol{\psi}_{t-1}, w_t^{\zeta}), \qquad \boldsymbol{\psi}_t \mid \boldsymbol{\psi}_{t-1} \sim \mathcal{N}(\boldsymbol{\psi}_{t-1}, \mathbf{W}_t^{\psi}).
$$

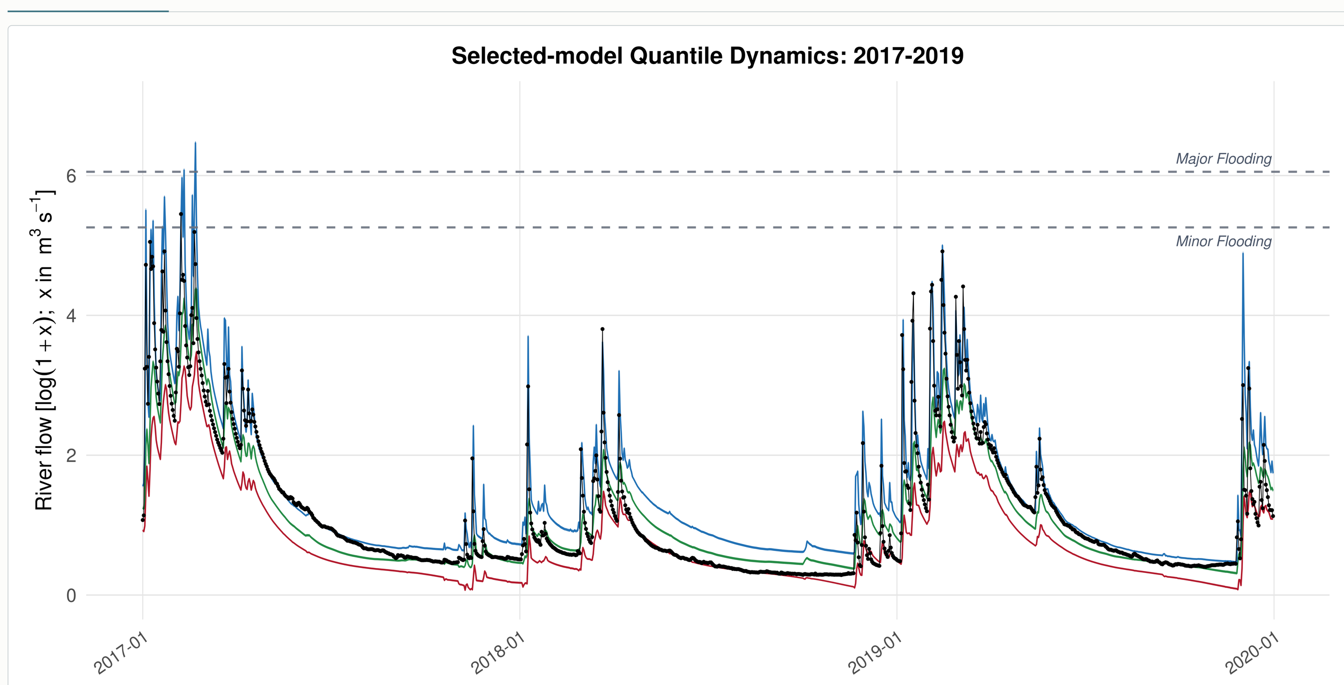
Here $\boldsymbol{\theta}_t$ is the shared trend–seasonal river quantile, $\boldsymbol{\delta}_t^j$ is the source-specific correction, and $(\zeta_t, \boldsymbol{\psi}_t)$ is the exogenous-covariate block driven by $\mathbf{x}_t$ (local precipitation, local soil moisture, and GDPC climate summaries). The DQLM comparator uses the conjugate AL likelihood; the selected exDQLM uses exAL and updates non-conjugate σ, γ terms with VB-LD.

RESULTS
1–28-step-ahead forecast CRPS comparison



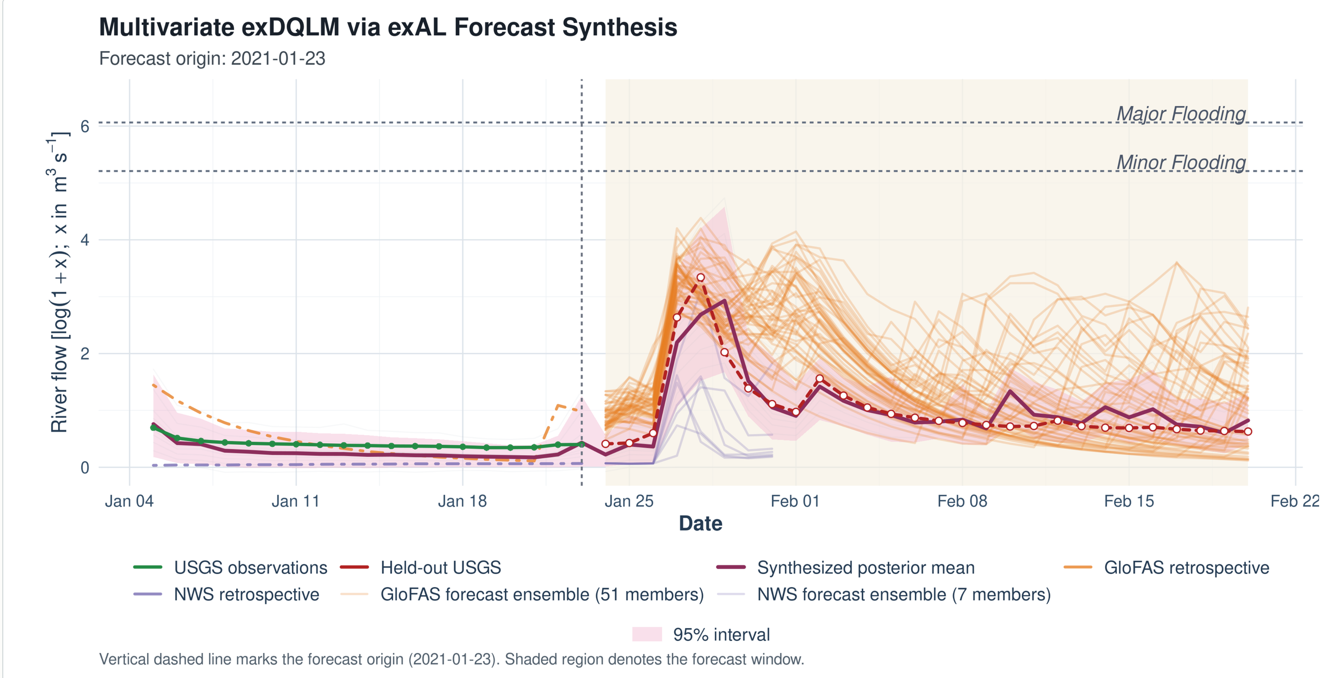
CRPS is the continuous ranked probability score; lower values are better. DQLM is the conjugate AL-likelihood baseline, while selected exDQLM denotes the exAL extension. Raw NWS is evaluated separately over common leads 1–8, not mixed into this 1–28 lead GloFAS-supported benchmark.

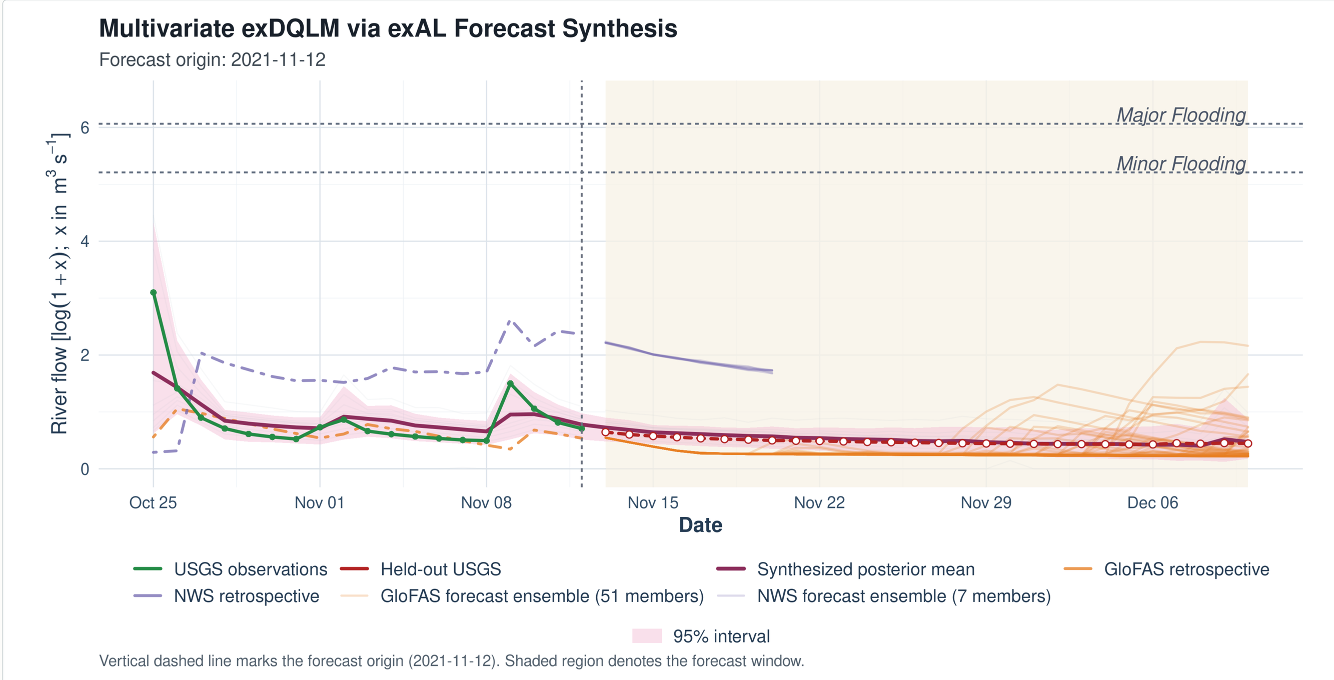
FIT
Wet-period quantile fit

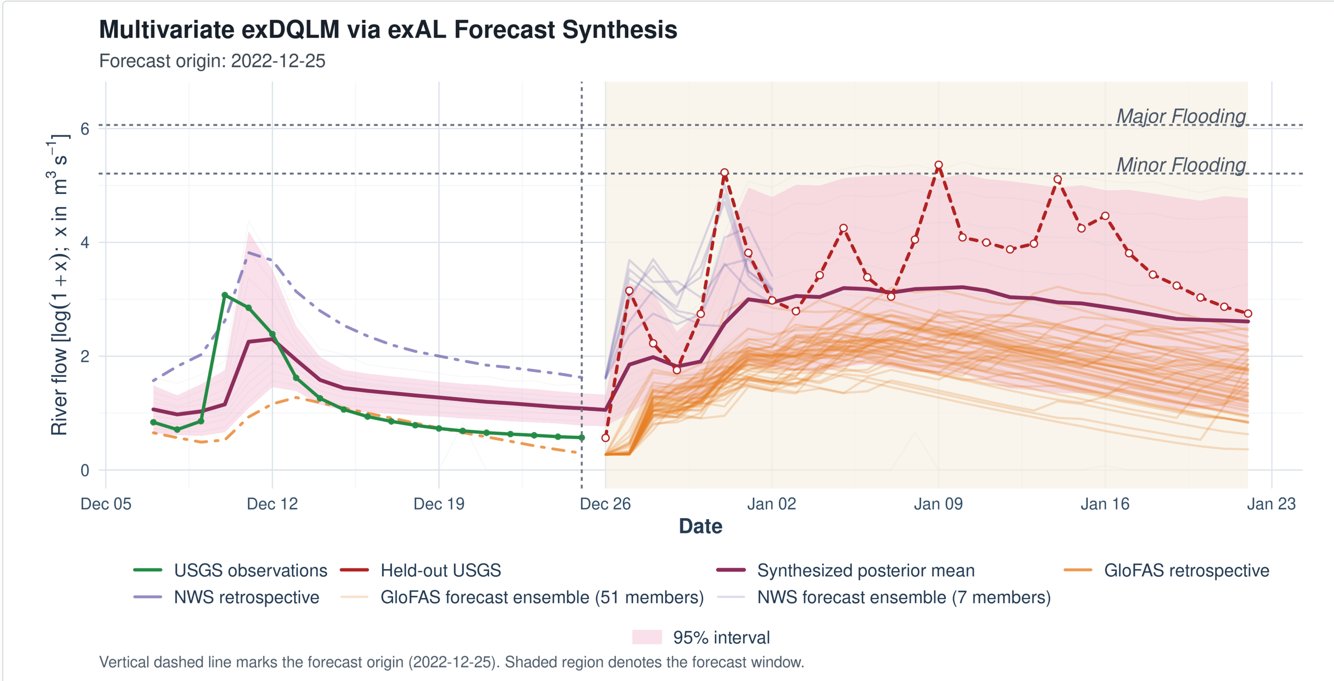


Historical fit diagnostic. Selected exDQLM fitted quantile locations during the 2017–2019 wet period. The 5th, 50th, and 95th quantile fits are shown with observed USGS flow; wider upper-tail bands make winter high-flow behavior visible.

EXAMPLE
Three forecast-window examples

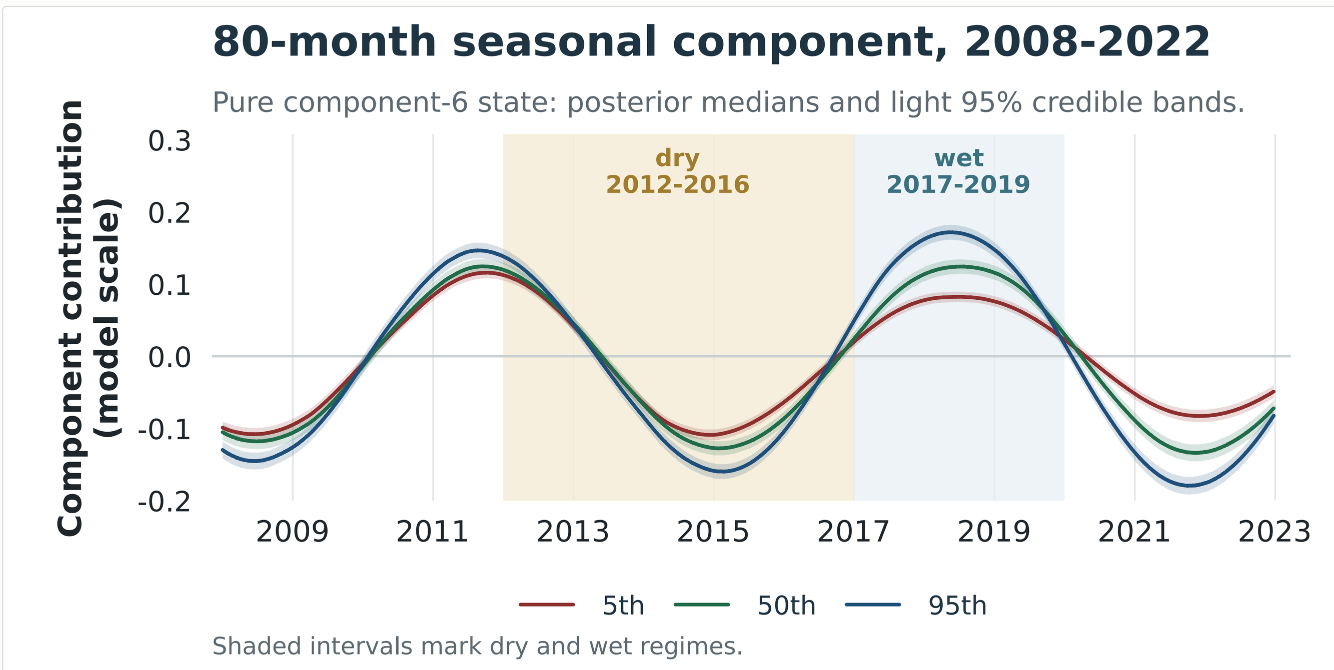






Origin-level illustrations. Three selected exDQLM forecast windows show how corrected quantile forecasts combine into a posterior predictive envelope after the cutoff. The panels are illustrative checks of behavior at individual origins; the aggregate comparison remains the five-origin CRPS benchmark.

DYNAMICS
Interpretable seasonal component



Retrospective diagnostic. One component of the interpretable state decomposition. From the selected exDQLM at the 2022-12-25 origin, lines show the isolated 80-month seasonal state, without the trend component added, for the 5th, 50th, and 95th target-quantile lanes over 2008–2022; light ribbons show 95% credible bands and shaded intervals mark dry and wet regimes.

TAKEAWAYS
Main takeaways

- Selected exDQLM has the lowest 1–28-step-ahead CRPS at four of five origins; DQLM leads at 2022-12-25.
- The model estimates quantile-level source bias from archived retrospectives, then propagates corrections into newly issued ensembles.
- Seven corrected quantile lanes synthesize one posterior predictive distribution on operational update timescales.
- The state decomposition separates source corrections, trend, seasonality, and exogenous effects; the 80-month panel shows one diagnostic component.

Code and reproducibility
github.com/AntonioAPDL/Project1